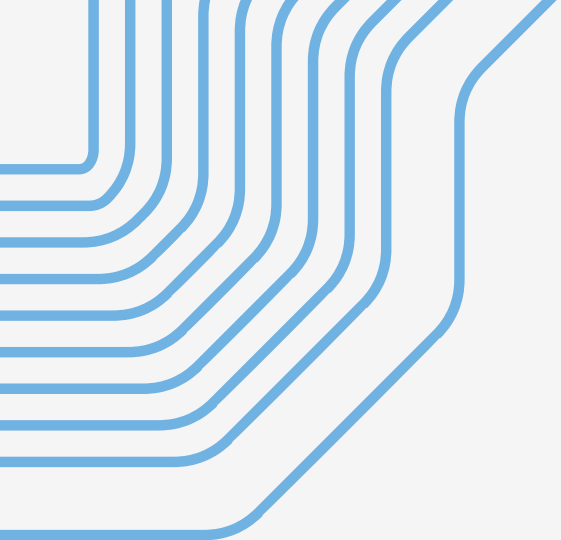


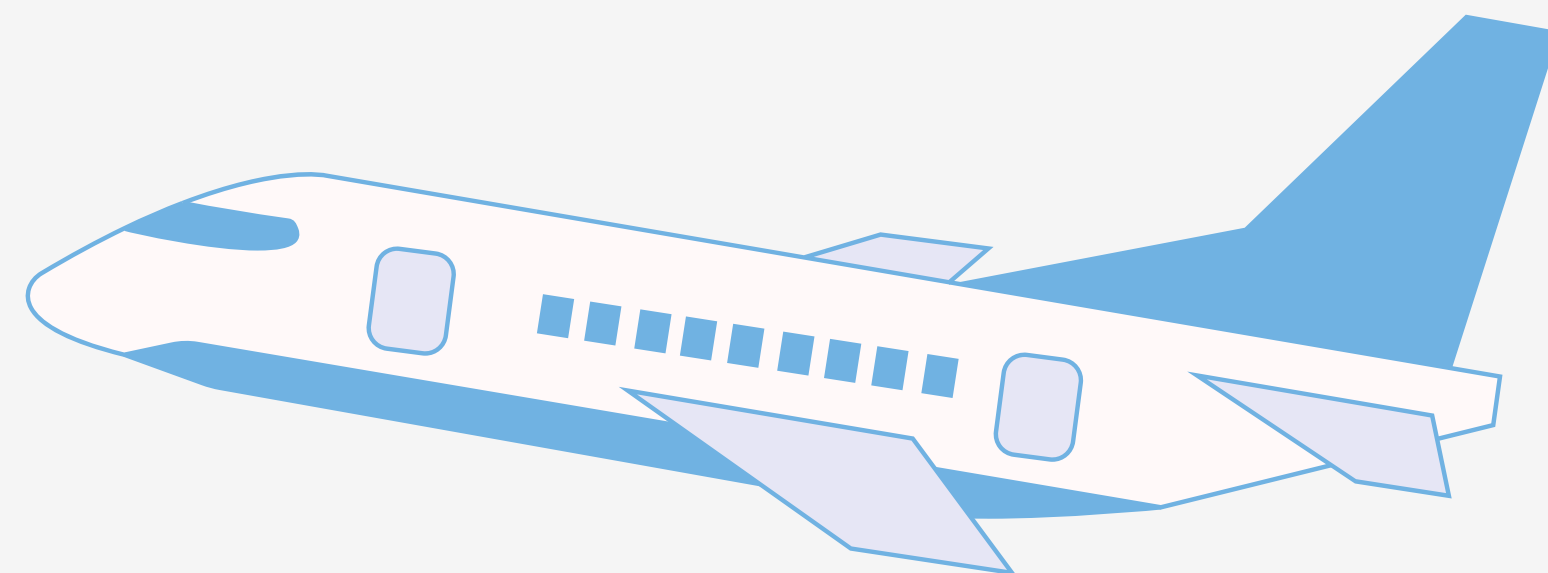


EUROPEAN AIR TRAFFIC

PERFORMANCE & DELAY ANALYSIS PIPELINE

Ekta Mistry

ETL & Pipeline Orchestration
ESILV MSc A4



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USE CASE & PROBLEM STATEMENT

How can we analyze in real-time the operational performance of European airports and identify delay patterns across the air network?

- Millions of flights per year in Europe
- Delays cost airlines ~100€/minute
- No simple open-source tool to monitor airport performance

Solution:

A complete ETL/ELT pipeline from raw GPS data to a live operational dashboard



PIPELINE ARCHITECTURE

2 Data Sources :

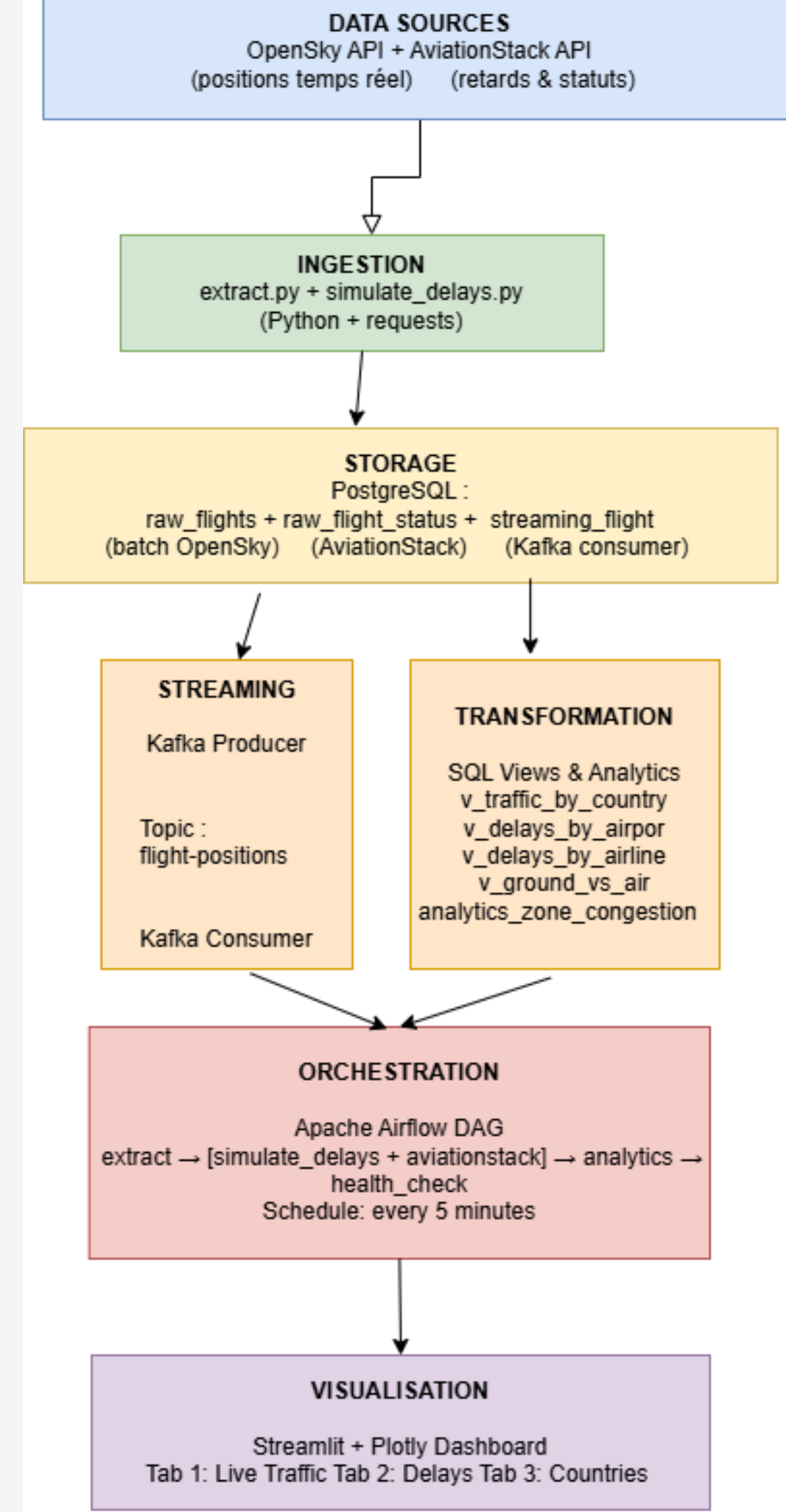
- OpenSky Network (live positions)
- AviationStack (delays & schedules)

5 layers :

- Ingestion
- Storage
- Streaming
- Transformation
- Visualisation

1 Orchestrator :

- Apache Airflow (every 5 minutes)



DATA & PIPELINE DETAILS

1. Data Sources

Live GPS positions of 1500+ European flights every 2 minutes. Flight schedules, real delays and statuses every hour for 8 major airports.

2. Ingestion Layer (ETL Batch)

Calls both APIs, cleans and filters data (removes null GPS positions), loads into PostgreSQL with idempotent inserts (ON CONFLICT DO NOTHING). Also generates realistic simulated delays as a fallback.

3. Storage Layer

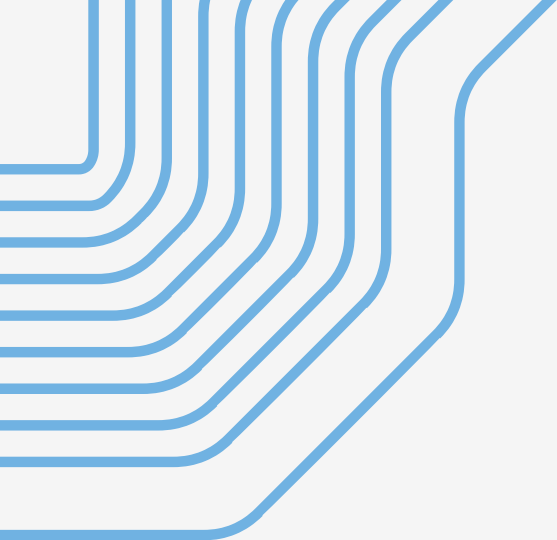
raw_flights : OpenSky positions
raw_flight_status : Delays & statuses
streaming_flights : Kafka consumer output

4. Streaming Layer

Producer reads from PostgreSQL every 60s and pushes to Kafka topic flight-positions. Consumer listens and writes to streaming_flights in real-time.

5. Transformation Layer

5 SQL views on top of raw tables: v_traffic_by_country, v_traffic_per_hour, v_ground_vs_air, v_delays_by_airport, v_delays_by_airline + analytics_zone_congestion table.



DASHBOARD & RESULTS

Live Traffic

- Real-time flight map over Europe
- Airport saturation (planes on ground)
- French traffic evolution (last 24h)

Delay Analysis

- Average delay by airport
- Top airlines by delay
- Flight status distribution

Country Analysis

- Top 15 countries by traffic volume
 - Ground vs airborne (Kafka stream)
- 